LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Methods and applications of electrical temperature indicating systems
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Methods and applications of electrical temperature indicating systems
5. Specific Objectives	15.1.5.1 Trainee will be able to explain various method of temperature indication system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of RTD and Thermocouple probe installation; Safety: High voltage precautions; Supervision: Instructor monitors connections and probe handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a temperature indicating system?
(b) Name two common types of electrical temperature sensors used in aircraft.
(c) Why is accurate temperature monitoring critical for engine performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Resistance Temperature Detector (RTD) principles • Thermocouple operation and Seebeck effect • Thermistor characteristics and applications • Signal conditioning and transmission methods	L & Ds	PPT, IB	15 min
(b)	• Comparison of RTD, Thermocouple, and Thermistor • Typical aircraft applications for each sensor type • Calibration and maintenance requirements • Fault detection and troubleshooting indicators	L & Ds	PPT, PDF	20 min
(c)	• Integration with aircraft warning systems • Data logging and remote indication techniques • Impact of electrical noise on sensor accuracy • Future trends in digital temperature sensing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference in output signal between an RTD and a thermocouple.
(b) Describe a scenario where a thermocouple would be preferred over an RTD in an aircraft engine.
(c) How does electrical noise affect the accuracy of a temperature indicating system?

12. Home Assignment and Follow Up	Prepare a brief note comparing the Ratio meter system and thermoelectric thermometers to be discussed in Lesson Plan 12.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Ratio meter system, thermoelectric thermometers	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min